Gebregziabihier Nigusie Birhane

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Education

Master of Science, Information Technology, Bahir Dar University, Ethiopia **Nov 2020 – Aug 2022**

- Thesis Title: Lexical Complexity Detection and Simplification in Amharic Text Using Machine Learning Approach.
- Honors: Best MSc Thesis Award, First-ranked Thesis in the Department Cohort.

Bachelor of Science , Information Technology, Mizan-Tepi University, Ethiopia. **Oct 2015 – Aug 2019**

- Project: Android Based Tepi Town Crime Report System.

Skills

Programming Languages

- Python is my primary language for AI/ML research, covering model training, evaluation, and data pipelines.
- TypeScript for front-end development, used to build the Heritage in Code project interface.
- LaTeX for document preparation, including scientific write-ups.
- Android and Flutter for mobile app development, including a pneumonia detection app from chest X-ray images (demo link).

ML Tools & Frameworks

- I used PyTorch and TensorFlow for ML training and GPU management.
- Hugging Face is used for fine-tuning pretrained language models and hosting deployed models, including the fusion engine model for the Heritage in Code project.
- FastAPI is used to serve deployed models through APIs and connect them to user-facing applications.

Databases

- MYSQL AND MONGODB I used them for structured project data storage and retrieval in most of my data-based projects, including the Heritage In Code project.

Git & GitHub

- I used Git for version control and GitHub for collaboration, pushing, pulling, and managing shared repositories for projects including Heritage in Code and Financial Agent.

Projects

Heritage in Code, Wits Innovation Center, South Africa – AI Engineer **Nov 2025 – Jun 2026**

- **Archival Pipeline:** Archived African heritage sounds alongside their cultural context and use.
- **Fusion Engine:** Developed a fusion engine for creative heritage sound reuse, using Chromaprint for audio similarity tracking and Demucs for source separation and pitch analysis.
- **Generative Baseline:** Used MusicGen as a baseline to benchmark heritage in code model's generative outputs. Project details: `https://www.heritageincode.com`

Financial Report Generation Agent, Rubani – AI Engineer **Mar 2026 – Present**

- **SQL Generation and Data Retrieval:** Developed a RAG pipeline for vector-based schema ranking and SQL generation from user prompts, retrieving relevant data from relational databases.

Projects (continued)

- **Report Generation:** Fine-tuned the Qwen-3B model, using RAG-based supervision, to synthesize retrieved tabular records into structured financial reports.

Research Experience

Graduate Researcher – Master's Thesis, Bahir Dar University, Ethiopia **Sep 2021 – Aug 2022**

- Fine-tuned pretrained language models, such as BERT and RoBERTa, for lexical complexity prediction and simplification in Amharic educational texts.
- Developed foundational complexity detection and simplification models with promising performance.
- Honors: Best MSc Thesis Award, ranked first in the department cohort.

Independent Researcher, Information Technology, Mizan-Tepi University, Ethiopia **Sep 2022 – Present**

- Extended thesis work on Amharic lexical complexity by validating the classification approach and refining the annotation methodology.
- Developed a contour-guided lung segmentation algorithm to validate chest X-ray images, and trained a lightweight mobile deep learning model for pneumonia detection, achieving real-time inference on mobile hardware for low-resource clinical deployment (MICCAI 2025).
- Analyzed sentiment polarity in Amharic-language social media discourse on climate change using LLM, presenting findings at Deep Learning Indaba 2025 and publishing results in PMLR 2026.

Publications

Nigusie, G., Elbatel, M., & Ayana, Y. (2026). Contour-guided segmentation and X-ray image validation for pneumonia detection using mobile deep learning in low-resource settings. In *International Workshop on Medical Image Computing in Resource Constrained Settings (MICCAI 2025)* (pp. 299–309). https://doi.org/10.1007/978-3-032-13654-1_30
- Presented at MICCAI 2025, Daejeon, South Korea.

Nigusie, G., Mossa, N., & Tegegne, T. (2026). Sentiment polarity analysis of Amharic climate change discourse using large language models. *Proceedings of Machine Learning Research*, 1–10. https://proceedings.mlr.press/v302/nigusie26a.html
- Presented at Deep Learning Indaba 2025, Kigali, Rwanda.

Nigusie, G., & Tegegne, T. (2023). Amharic text complexity classification using supervised machine learning. In *Artificial Intelligence and Digitalization for Sustainable Development (ICAST 2022)* (pp. 12–23). https://doi.org/10.1007/978-3-031-28725-1_1
- Presented at CAST 2022, Bahir Dar, Ethiopia.

Nigusie, G., & Tegegne, T. (2022). Lexical complexity detection and simplification in Amharic text using machine learning approach. In *IEEE 4th International Conference on ICT for Development for Africa (ICT4DA 2022)* (pp. 1–6). https://doi.org/10.1109/ICT4DA56482.2022.9971306

Awards & Grants

- AI and African Music Project Grant ($5,500), Wits University, South Africa, November 2025.
- MICCAI Society MIRASOL Travel Grant (€3,500), Daejeon, South Korea, September 2025.
- Best MSc Thesis Award, Bahir Dar University, Ethiopia, August 2022.

Volunteer

Mentor, Deep Learning Indaba 2024 Mentorship Program **Sep 2024 – Jan 2025**

Volunteer (continued)

- Mentored a team of four students project, providing technical guidance on AI model implementation.

References

Available on Request